

## ASSIGNMENT: Build an Advanced Document Q&A RAG System with Multi-Agent Architecture

### OVERVIEW:

Create a sophisticated RAG (Retrieval-Augmented Generation) application where users upload documents (text/PDF), build a knowledge base and then query the content. The system uses a multi-agent architecture with MCP (Model Context Protocol) integration to retrieve relevant info, perform web searches when needed, and generate accurate answers using an LLM.

MUST USE: LangGraph for agent orchestration

### CORE REQUIREMENTS

#### 1. MULTI-AGENT ARCHITECTURE

Implement a coordinated multi-agent system with the following agents:

Orchestrator Agent, Query Analysis Agent, Retrieval Agent, Re-ranking Agent, Generation Agent, Citation Agent

#### 2. MCP (MODEL CONTEXT PROTOCOL) INTEGRATION

Required MCP Servers:

- Web Search MCP Server, Vector Database MCP Server, Document Processing MCP Server (Optional but Recommended)

#### 3. DOCUMENT PROCESSING

- Accept text files (.txt, .md) and PDFs
- Extract text content accurately
- Support image extraction from PDFs with OCR
- Chunk documents intelligently (consider semantic boundaries)
- Generate embeddings for all chunks
- Preserve document metadata (filename, page numbers, timestamps)

#### 4. HYBRID SEARCH IMPLEMENTATION

#### 5. CONVERSATION MEMORY & HISTORY

- Implement conversation history tracking
- Enable follow-up questions with context awareness
- Store conversation turns in memory
- Use conversation buffer or summary memory
- Allow users to reference previous questions/answers

#### 6. BACKEND & FRONTEND ARCHITECTURE

Backend (Fast API):

- Build RESTful API using Fast API
- Appropriate Endpoints as required

Frontend (HTML/CSS/JavaScript):

- Build single-page application using vanilla HTML, CSS, and JavaScript
- No frameworks required (React, Vue, etc.) - pure HTML/CSS/JS only
- Can use simple libraries like Bootstrap for styling (optional)

#### DELIVERABLES

1. Source code repository

2. Documentation

3. Demo:

- Sample documents for testing
- Example queries demonstrating key features